

Customer Behaviour and Retail Analytics

1. Executive Summary

Understanding customer purchasing behaviour is essential for retailers aiming to improve sales performance and customer retention. This project examines transactional retail data to identify spending trends, customer preferences, loyalty patterns, and factors influencing purchasing decisions.

Using Python for data preparation, MySQL for analytical querying, and Power BI for visualization, the project converts raw transactional data into actionable business insights that support strategic decision-making.

2. Project Objective

The primary objective of this project is to analyze customer purchase behavior and determine how demographic attributes, subscription programs, promotional activities, and shopping preferences impact business performance.

The analysis seeks to answer the following business question:

How can retail organizations use customer transaction data to enhance customer engagement, increase revenue, and improve retention strategies?

3. Dataset Description

The dataset contains retail transaction records collected from customer purchases across multiple product categories.

Dataset Characteristics

- Total Transactions: 3,900
- Total Features: 18
- Dataset Type: Retail Customer Transactions

Information Available

- Customer demographics
- Product and category details
- Purchase amounts

- Seasonal buying patterns
- Subscription membership
- Product reviews and ratings
- Discount utilization
- Shipping preferences

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	NaN
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	NaN
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

Data Quality

A small number of missing values were identified within the review rating attribute and addressed during preprocessing.

4. Data Preparation and Transformation (Python)

Data preparation was performed using Python to ensure accuracy and consistency before analysis.

Activities Performed

- Imported and inspected raw data using Pandas.
- Evaluated dataset structure and summary statistics.
- Identified and treated missing values.
- Standardized column naming conventions.
- Created customer age segments for behavioral analysis.
- Generated additional analytical fields to support reporting.
- Removed duplicate or redundant information where appropriate.
- Validated dataset quality before database integration.

Database Loading

After preprocessing, the cleaned dataset was loaded into MySQL to support structured business analysis and reporting.

5. Business Analysis Using MySQL

MySQL queries were developed to explore customer behavior, spending habits, and revenue drivers.

Analysis Areas

Revenue Contribution by Gender

Evaluated revenue generated by different customer groups to identify valuable market segments.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

High-Value Customers Using Discounts

Identified customers who continued to spend above average despite utilizing promotional offers.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:00

Product Satisfaction Analysis

Determined products receiving the highest customer ratings and strongest overall feedback.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Shipping Preference Impact

Compared purchase values between customers selecting standard and premium shipping options.

Subscription Performance Analysis

Measured differences in spending and revenue contribution between subscribers and non-subscribers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Discount-Oriented Product Analysis

Examined products most frequently purchased using discounts and promotional incentives.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Customer Segmentation

Grouped customers into behavioral categories such as New, Returning, and Loyal customers based on purchasing history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Product Popularity Assessment

Identified the highest-performing products within individual product categories.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Loyalty and Subscription Relationship

Analyzed whether repeat customers demonstrate stronger adoption of subscription services.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Revenue Distribution by Age Group

Measured revenue age segments.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

6. Dashboard Development (Power BI)

A Power BI dashboard was developed to communicate findings through interactive visualizations.



Dashboard Features

- Customer Segmentation Overview
- Revenue Performance Metrics
- Subscription Impact Analysis
- Product Performance Tracking
- Shipping Preference Insights
- Customer Spending Patterns

The dashboard enables stakeholders to quickly identify trends and monitor business performance.

7. Key Findings

Customer Spending Trends

Customer purchasing behavior varies across demographic groups, indicating opportunities for targeted marketing initiatives.

Subscription Benefits

Subscription members contribute a significant share of revenue and generally demonstrate stronger purchasing activity.

Premium Shipping Preference

Customers selecting express delivery options tend to exhibit higher transaction values.

Customer Loyalty Impact

A relatively small segment of loyal customers contributes a substantial portion of total revenue.

Product Performance

Certain products consistently achieve stronger customer satisfaction ratings and purchase frequency.

Promotion Effectiveness

Discount programs encourage purchases but require careful monitoring to maintain profitability.

8. Business Recommendations

Expand Subscription Programs

Introduce additional membership benefits to encourage long-term customer retention.

Strengthen Loyalty Initiatives

Reward repeat customers through personalized incentives and loyalty-based rewards.

Implement Targeted Marketing

Use customer segmentation insights to develop focused promotional campaigns.

Promote High-Performing Products

Prioritize highly rated products within advertising and marketing strategies.

Optimize Delivery Services

Encourage premium shipping options through targeted offers and improved customer experience.

Monitor Discount Performance

Evaluate promotional effectiveness regularly to balance revenue growth and profit margins.

9. Conclusion

This project demonstrates how retail transaction data can be transformed into valuable business intelligence through data cleaning, database analysis, and visualization.

The insights generated help organizations better understand customer behavior, improve retention strategies, and make informed business decisions. By leveraging customer analytics, retailers can enhance customer experiences while driving sustainable revenue growth.